5 Fig. 5.1 shows a point source S emitting continuous waves at a distance *h* from a plane reflector R. As a result, regions of high intensity such as C, C' and C" are produced.

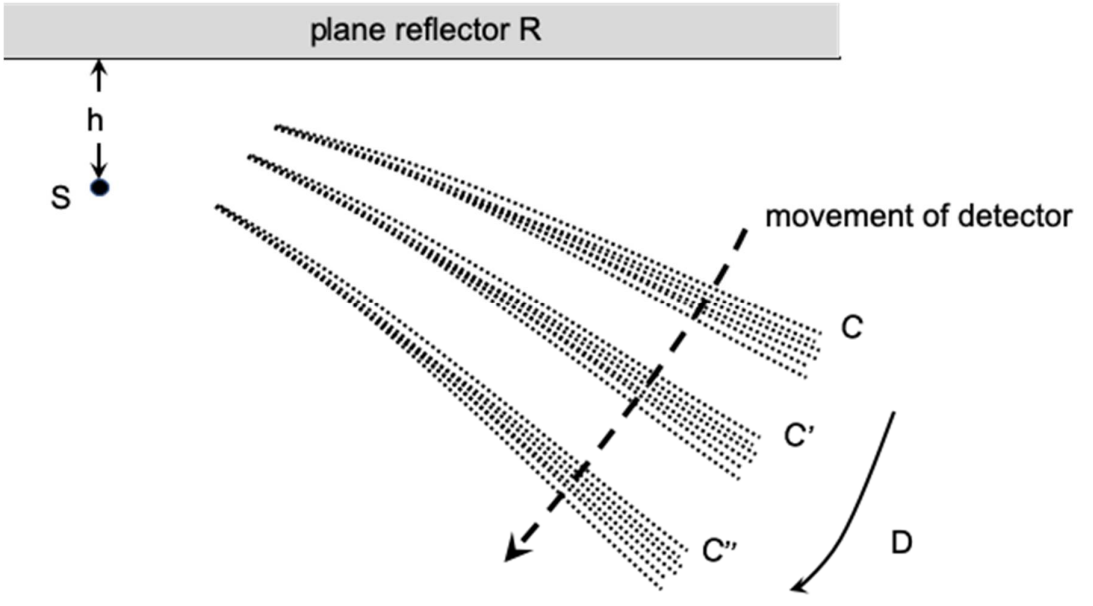


Fig. 5.1

(a) Explain why regions of high intensity such as C, C' and C'' are produced.

[3]

(b) A detector is moved in the direction indicated by the dotted arrow in Fig. 5.1. Explain how the detected intensity of the waves would vary.

[3]

(c) When the frequency of S is decreased slowly, the regions C, C' and C'' move in the direction D as shown in Fig. 5.1. Explain this phenomenon.

[2]